24CC72

Third Semester B.Com. Degree Examination, December/January 2025-26

(SEP Syllabus)

Business Statistics

Time : 3 Hours

Max. Marks : 80

SECTION – A

I. Answer **any ten** of the following sub-questions. Each sub-question carries **2** marks : **(10 × 2 = 20)**

1. (a) Define Tabulation.

   (b) What is collection of data?

   (c) Define Statistics.

   (d) What is classification?

   (e) What is frequency?

   (f) Define correlation.

   (g) What is Regression?

   (h) Mention types of measures of central tendency.

   (i) What is Median?

   (j) What is secondary data?

   (k) What is probable error?

   (l) What is Ogives?

SECTION – B

II. Answer **any six** of the following. Each question carries **5** marks : **(6 × 5 = 30)**

2. Explain the difference between primary data and secondary data.

3. Explain the functions and limitations of Statistics.

4. Draw a Blank Table to show the number of students in a college classified according to its faculty.
   (a) Faculty : Arts, Science, Commerce
   (b) Age Groups : Below 18 years, 18-20 years, above 20 years
   (c) Gender : male, female

5. Construct a Frequency Table of marks obtained by 50 students in exclusive classes as given below taking class interval of 10 :

42	53	65	63	61	47	56	60	64	45
55	57	82	42	39	51	65	55	33	70
50	52	53	45	45	25	36	59	63	39
65	30	45	35	49	15	54	48	64	26
75	20	42	40	41	55	52	46	35	18

6. Draw a Histogram and locate mode graphically from the following data. Verify the result by formula.

Marks :	0–10	10–20	20–30	30–40	40–50
Frequency :	5	12	22	10	3

7. Calculate median from the following data :

Marks :	10	20	30	40	50	60	70	80	90
No. of students :	3	5	9	3	2	3	5	10	10

8. Find the Range and the coefficient of Range for the following data :
   65, 70, 82, 59, 81, 76, 57, 60, 55 and 50.

9. Calculate the standard deviation and coefficient of variation.
   X : 10, 15, 20, 30, 35, 32, 20, 5.

SECTION – C

III. Answer **any three** of the following. Each question carries **10** marks : **(3 × 10 = 30)**

10. Explain the differences between correlation and regression.

11. Calculate the coefficient of correlation between the age of father and son from the following data :

		Father			
		0–10	10–20	20–30	30–40
Son	10–20	5	4	3	–
	20–30	7	6	7	6
	30–40	–	5	–	7

12. Compute mean, median and mode from the following data :

Marks	No. of students
0–10	7
10–20	9
20–30	18
30–40	36
40–50	54
50–60	75
60–70	51
70–80	27
80–90	15
90–100	8

13. Calculate coefficient of correlation between Investment made and return obtained from the following data. Assume 105 and 60 as mean value of investment made and return obtained and also calculate probable error of 'r'.

Investment made :	105	111	104	112	118	98	116	103	116	112
Return obtained :	62	64	53	60	72	56	68	60	69	65

14. From the following data obtain the Regression of Y on X and X on Y. Estimate the Y when X is 30.

X :	25	28	35	32	36	29	32	38	34	32
Y :	43	46	49	42	36	32	31	30	33	39